

From Radio Waves to Wellness: Radar and Machine-Learning for Non-Contact De-Hydration Assessment

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Abstract—This paper introduces a novel, noninvasive, and contactless method for dehydration measurement using frequency-modulated continuous wave (FMCW) radar operating at center frequencies of 10 GHz and 77 GHz. The proposed framework begins with a customized preprocessing technique designed to isolate the signal corresponding to the target of interest (ToI) from the received radar data, ensuring that the extracted features reliably reflect physiological indicators of hydration status. To capture the complex, nonlinear patterns associated with dehydration, we employ a wavelet scattering transform (WST)-based feature extraction method, which provides robust inputs for machine learning models. These features are then used to train and evaluate various classifiers, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Naive Bayes, and Neural Networks, to accurately determine hydration levels. To overcome the challenge of limited labeled data in radar-based dehydration monitoring, we developed and annotated a novel dataset—the *Fasting Dataset*—collected under controlled conditions from individuals observing fasting. This dataset captures radar measurements at different physiological stages, offering a valuable benchmark for hydration-level analysis. Validation using this custom dataset demonstrates the strong potential of the proposed methodology for real-time, noninvasive, and contactless dehydration monitoring in remote healthcare environments.

Index Terms—Dehydration monitoring, non-contact, FMCW Radar, Wavelet Scattering Transform

I. INTRODUCTION

The COVID-19 pandemic and the growing aging population have significantly increased the demand for non-invasive and contactless sensing technologies in healthcare monitoring within hospitals and nursing homes [1], [2], [3]. These advanced sensing methods are crucial for continuous health monitoring, reducing the risk of infection transmission, and improving patient comfort and care. Non-contact sensing technologies are being leveraged for a variety of critical applications, including vital sign estimation, gesture recognition, road user classifications, emotion recognition, speech recognition and sleep monitoring, etc. To our knowledge, the main non-contact sensing technologies are radio frequency sensing, vision sensing, and acoustic sensing [4]. Radio frequency

signals for contactless sensing include WiFi, ultra-wideband (UWB), and millimeter-wave (mmWave) technologies. Vision-based technologies are based on visible light and infrared (IR), while acoustic methods typically involve the use of audio devices for signal acquisition or extraction. These non-contact sensing modalities offer unique advantages and have been integrated into various healthcare and non-healthcare applications, demonstrating their versatility and effectiveness.

Beyond monitoring vital signs, falling detection, and depression, monitoring dehydration levels is crucial in healthcare. Dehydration occurs when the body loses more fluid than it takes in, affecting normal function. It can happen on hot days with excessive sweating or from not drinking enough water. Dehydration is a serious health problem, especially in vulnerable groups such as the elderly, infants, and those with chronic diseases. Early detection and continuous monitoring are essential to prevent severe complications and ensure timely treatment. Effective monitoring of dehydration in healthcare care can significantly improve patient care, reduce hospital admissions, and improve overall health outcomes. Traditional methods of measuring dehydration include assessing physical signs such as dry mouth, rapid heartbeat, and low blood pressure. Blood tests can measure electrolyte levels, blood urea nitrogen (BUN), and serum osmolality, offering precise information on the hydration status of a patient. Urine analysis involves monitoring urine output and specific gravity, which also indicate hydration levels. However, these methods are invasive, uncomfortable, and not ideal for continuous monitoring.

Noninvasive dehydration monitoring methods often use wearable sensors, such as oximeters, smartwatches, and wristbands to capture photoplethysmography (PPG) and electrodermal activity (EDA) signals, which are then analyzed using machine learning (ML) algorithms [5]. For example, [6] used EDA and PPG data from 17 subjects to detect mild dehydration through autonomic responses to cognitive stress, achieving notable results. The authors in [7] used EDA data from 16 subjects in different postures to classify hydration levels with a hybrid Bi-LSTM neural network. The work in [8] used PPG data from 17 dehydrated patients in an emergency setting, achieving 67.91% precision with an SVM classifier. Similarly, the work in [9] combined data on EDA, skin temperature, heart rate, and BMI of 16 participants during physical activity to quantify fluid loss. A real-time Android tool was developed using EDA data in [10], achieving an accuracy of 84.5%.

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Similarly, [11] used EDA data from five subjects in different activities and achieved 91.3% accuracy with a random forest classifier. In another study [12] reported an accuracy of 87.78% for the detection of dehydration using a k -NN classifier with EDA data. [13] collected 1-lead electrocardiogram (ECG) data from fasting and sportspeople subjects, and implemented a number of ML, deep learning (DL) and large language models (LLM) for non-invasive dehydration monitoring. Additionally, due to the prevalence of smartphones in today's world, there have been attempts to utilize the various sensors of the smartphone in order to turn it into a dehydration diagnosis tool, e.g., [14] utilized the camera of the smartphone to obtain a short video of the fingertip in order to extract PPG signal out of it which was then utilized to infer the dehydration level of a person. Similarly, [15] proposed to utilize the capacitive touchscreen of the smartphone for measurement of skin hydration level of a person. Finally, [16] took a different approach using data from the leg skin microbiome of 63 female subjects to predict skin hydration levels and other biomarkers.

Going beyond non-invasive (i.e., contact-based) methods, non-contact health monitoring technologies have the potential to revolutionize dehydration detection and management [17]. These technologies have the potential to deliver continuous, accurate, and unobtrusive monitoring—enhancing patient comfort and compliance while simultaneously reducing the burden on clinical staff. Such systems could be seamlessly integrated into hospitals, nursing homes, and remote care settings, paving the way for more proactive and personalized care. As the demand for advanced healthcare solutions grows, developing and implementing non-contact dehydration monitoring technologies is a significant step forward.

Among the emerging non-contact technologies, radar stands out as a particularly promising candidate. While traditionally employed in military and aerospace applications for object detection and localization, radar is now gaining momentum in civilian domains such as healthcare and autonomous systems [18], [19], [20]. Notably, mmWave radar offers accurate monitoring of both physical and physiological signals, owing to its use of non-ionizing radiation, low cost, high resolution, low power consumption, and robustness in challenging environments such as fog and rain [21], [22], [23]. This paper presents a preliminary investigation into the feasibility and potential of radar as a non-contact tool for dehydration assessment.

The **main contribution** of this paper are as follows:

- We present the first radar and machine learning-based system for non-invasive, contactless dehydration monitoring—offering a novel and practical alternative to traditional assessment methods. This approach holds significant promise for monitoring hydration levels in challenging cases, including individuals who are seriously injured or burned, mentally ill, or infected.
- To address the scarcity of labeled data for radar-based dehydration detection, we developed a custom dataset under controlled conditions using fasting individuals. The Fasting Dataset captures radar signals at multiple stages

of fasting, systematically labeled into five phases of dehydration.

- We apply targeted preprocessing techniques to isolate the target-of-interest (ToI) from the received radar signals. This ensures that the extracted features robustly reflect the subject's physiological state, significantly improving the system's reliability and accuracy.
- For feature extraction, we utilize the Wavelet Scattering Transform (WST), which effectively captures intricate and nonlinear signal characteristics. These rich features serve as inputs to machine learning models for accurate classification of dehydration levels.

The overall paper is organized as follows: Section II discusses the relationship between dehydration and skin permittivity. Section III presents the signal model developed for dehydration monitoring. Section IV outlines the experimental setup and the methodology used for signal acquisition and processing. Section V describes the classification strategies employed to distinguish between different stages of dehydration. The experimental results are presented in Section VI, and concluding remarks are provided in Section VII.

II. SKIN DEHYDRATION INSIGHTS FOR RADIO WAVES

The hydration state of a patient can be monitored by measuring the water content in the skin, a multi-layered structure whose dielectric properties reflect its hydration level. As depicted in Fig. 1, human skin comprises distinct layers, each contributing to general permittivity. Crucially, the permittivity of the skin is not static; it varies with both the frequency of the incident signal and the water content of the tissue.

The frequency-dependent relative permittivity spectrum can be modeled using the Cole-Cole expression [24]:

$$\epsilon(\omega) = \epsilon_{\infty} + \sum_n \frac{\Delta\epsilon_n}{1 + (j\omega\tau_n)^{(1-\alpha_n)}} + \frac{\sigma_i}{j\omega\epsilon_o} \quad (1)$$

where the model parameters ϵ_{∞} represent the permittivity at high frequencies ($\omega\tau \ll 1$), and ϵ_s is the permittivity at low frequencies $\omega\tau \gg 1$. The term $\Delta\epsilon = \epsilon_s - \epsilon_{\infty}$ denotes the change in permittivity, while α is the distribution parameter, τ is the mean relaxation time, and σ is the static ionic conductivity. This relationship enables monitoring dehydration by probing the electrical permittivity of the skin, which can be compactly expressed as

$$\epsilon(\omega) = \epsilon'(\omega) - j\epsilon''(\omega)$$

where:

- $\epsilon'(\omega)$ (Real Part): Represents the material's ability to store electrical energy. Higher values indicate a stronger interaction with the electric field and greater energy storage, as observed in hydrated tissues or polar liquids. Lower values correspond to weaker energy storage, typical of low-loss materials such as rubber or air.
- $\epsilon''(\omega)$ (Imaginary Part): It characterizes the energy dissipation or loss within a material. Elevated values indicate higher absorption and conductive losses, as seen in biological tissues or aqueous solutions. In contrast, low values are typical of insulating materials (e.g., dry skin,

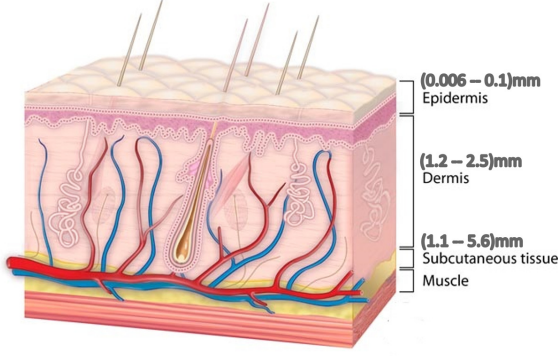


Fig. 1. Structure of the different layers of the skin: epidermis, dermis, and subcutaneous tissue. (Source: <https://dermnetnz.org/cme/principles/structure-of-the-epidermis>)

plastics). This term also explains why microwave ovens heat food effectively: water inside the food has a high ϵ'' value, enabling it to absorb microwave energy and convert it into heat. Furthermore, the same property governs how deeply electromagnetic waves can penetrate into materials, with greater absorption leading to shallower penetration.

In [25], the penetration depth (denoted by δ) of electromagnetic waveforms into biological tissues is analyzed using Maxwell's inhomogeneous equations. Specifically, the study derives the penetration depth for various organs, including dry skin. The computed values of δ for dry skin, when permittivity data are obtained from [26] at various frequencies, are summarized in Table I. To estimate the parameters required in (1),

f	ϵ'	$\sigma = \frac{\epsilon''}{\omega}$	Penetration depth ($\delta(f)$)
10 GHz	31.3	8.01	3.80 mm
24 GHz	19.00	22.8	1.10 mm
77 GHz	6.56	38.2	0.40 mm

TABLE I
RELATIVE PERMITTIVITY, CONDUCTIVITY LOSSES, AND PENETRATION DEPTH OF DRY SKIN AT DIFFERENT FREQUENCIES.

the work in [27] uses a comprehensive data set comprising 22 sources from the literature [28] as well as experimental measurements [29]. We used the database provided in [26] to extract the permittivity values of dry skin, tongue, and water for a comparative analysis of their dielectric properties. The resulting relative permittivity values are illustrated in Fig. 2, highlighting the differences in their electromagnetic behavior. Due to the dielectric properties of biological tissues, an electromagnetic signal interacting with the skin undergoes varying degrees of attenuation and phase shifts. These variations are primarily influenced by the skin's hydration level, as water content significantly affects the tissue's dielectric behavior.

To the best of our knowledge, radar signals have not yet been utilized for dehydration monitoring. In this study, our ob-

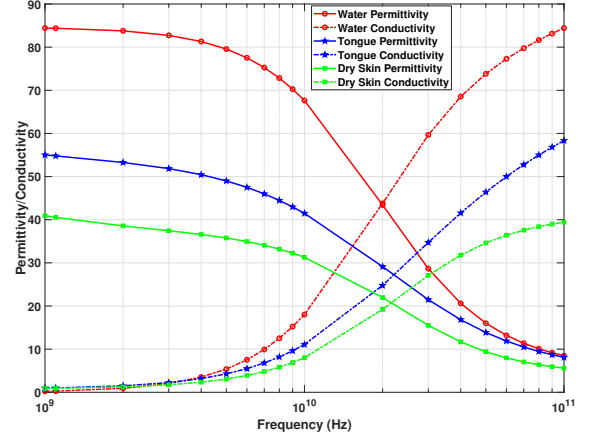


Fig. 2. Permittivity and conductivity ($\sigma = \frac{\epsilon''}{2\pi f}$) of dry skin, tongue, and water at different frequencies [24].

jective is to address this gap by leveraging radar data collected from real human subjects to classify and distinguish between different stages of dehydration. Our approach demonstrates the potential of radar as a novel, noninvasive, and noncontact tool for hydration assessment, with promising applications in medical, sports, and wellness settings.

III. SIGNAL MODEL

Consider a scenario, as shown in Fig. 3, in which the chest of a patient is exposed to electromagnetic radiation (EM) from a radar transmitting an FMCW signal.

$$x(t) = Ae^{j2\pi(f_c + \frac{Bt}{2T})t}, \quad (2)$$

where f_c is the carrier frequency of the radar, T is the pulse width, and B is the signal bandwidth. The frequency of the transmitted signal at time t , $f(t)$, can be estimated by differentiating the phase of the signal in (2).

As illustrated in Fig. 3, when the transmitted waveform interacts with the human body, part of it reflects off the surface, while the rest penetrates the chest and reflects from various internal layers. These reflections originate from different depths, resulting in a received signal composed of multiple frequency components, each with a distinct amplitude. The amplitude of each component is influenced by the body's water content, enabling the analysis of physiological changes based on the reflected signal. Consequently, the exposed body of the patient can be modeled as a dynamic channel, where the impulse response varies in response to fluctuations in the water content. Given this model and defining $h(t)$ as the body's impulse response, the signal after reflection from the human body at the surface of the chest can be expressed as

$$\begin{aligned} x_r(t) &= h(t) * x(t - t_o), \\ &= \int h(\tau) e^{j2\pi(f_c + \frac{Bt}{2T})(t - t_o - \tau)} d\tau. \end{aligned} \quad (3)$$

where t_o is the time it takes for a waveform to cross the radar to the surface of the body. It is important to note that the

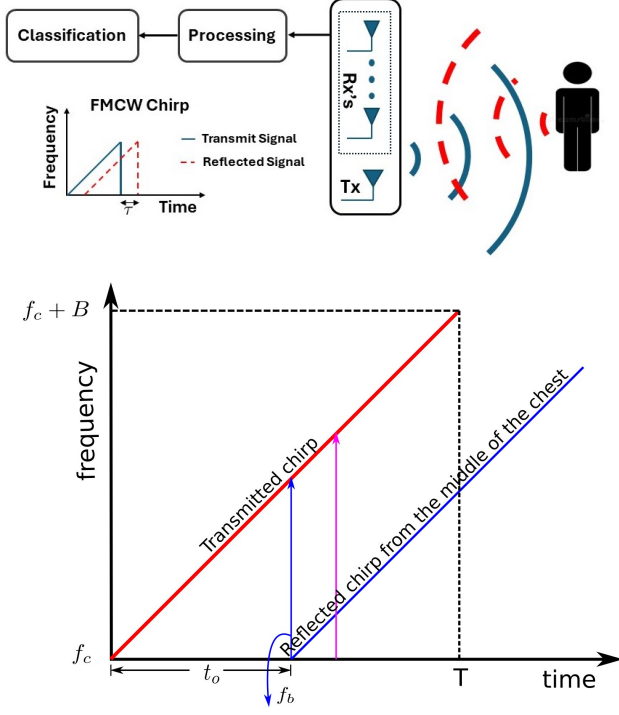


Fig. 3. The transmitted and reflected signals from the patient's chest will vary depending on the body's dehydration level, which acts as a channel. Different frequencies will experience varying levels of attenuation based on the hydration status.

above response holds true only if the body's impulse response, $h(t)$, remains consistent across all frequencies. However, in most practical scenarios, $h(t)$ may vary with frequency. In such cases, a time-frequency analysis is required to accurately capture the body's behavior.

In radar systems, where the transmitter and receiver are in one location, the received signal undergoes demodulation at the receiver. This process can be mathematically expressed as

$$\begin{aligned}
 z(t) &= x_r(t - t_o)x^*(t), \\
 &= \int h(\tau)e^{j2\pi(f_c + \frac{Bt}{2T})(t-2t_o-\tau)}e^{-j2\pi(f_c + \frac{Bt}{2T})t}d\tau, \\
 &= e^{-j2\pi(f_c + \frac{Bt}{2T})2t_o} \int h(\tau)e^{-j2\pi(f_c + \frac{Bt}{2T})\tau}d\tau, \\
 &= e^{-j2\pi(f_c + \frac{Bt}{2T})2t_o} H\left(f_c + \frac{Bt}{2T}\right) \\
 &= e^{-j\frac{4\pi R_o}{\lambda}} H\left(f_c + \frac{Bt}{2T}\right) e^{-j2\pi\frac{Bt_o}{T}t},
 \end{aligned}$$

where $H(f_c + \frac{Bt}{2T})$ represents the gain of the channel at frequency $(f_c + \frac{Bt}{2T})$, which is changing with respect to time. In this formulation, if the human body attenuates the reflected signal uniformly at any given time t , the demodulated signal at the receiver produces a constant beat frequency. However, fluctuations in the skin's water content can lead to variations in this attenuation.

To further illustrate this concept, we conducted a simulation. Fig. 4 and Fig. 5 present the results for an FMCW chirp signal reflected from two types of surfaces: a frequency-flat

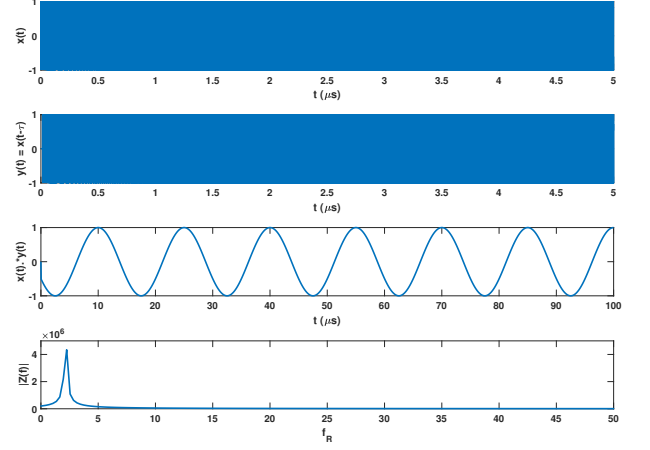


Fig. 4. Top to bottom: transmitted chirp signal, delayed reflected chirp from a frequency-flat surface located 2 meters away, demodulated received signal, and the corresponding scaled beat frequency.

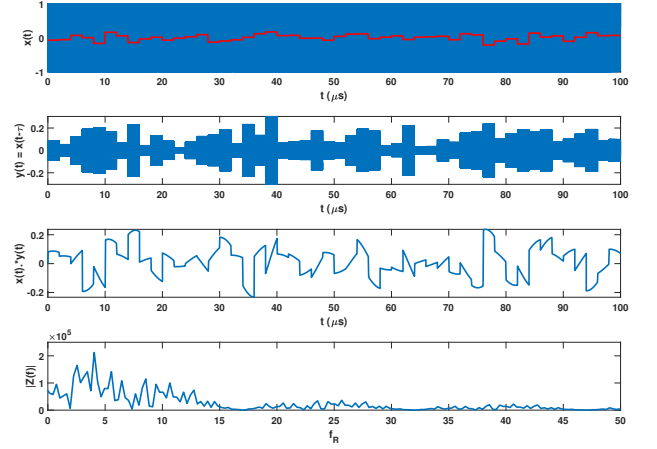


Fig. 5. Top to bottom: transmitted chirp signal, delayed reflected chirp from a frequency-selective surface located 2 meters away, demodulated received signal, and the corresponding scaled beat frequency.

surface and a frequency-selective surface. In the case of the frequency-flat surface, all frequency components are uniformly attenuated, resulting in a demodulated signal that exhibits a single dominant beat frequency, as shown in the third row of Fig. 4. To simulate a frequency-selective surface, the signal's bandwidth is divided into an arbitrary sub-bands, each of which is multiplied by a complex random coefficient to vary its amplitude and phase. When the FMCW signal reflects off such a surface, each sub-band experiences different levels of attenuation. As a result, the demodulated signal exhibits a spread spectrum rather than a single beat frequency, as illustrated in the fourth row of Fig. 5.

This spectral variation captures the surface's frequency response, which, in the case of biological tissues, can be associated with water content. Thus, the frequency-selective behavior introduces distinct spectral signatures into the received signal, providing a promising avenue for characterizing

tissue hydration.

Building on this, the received signal can be further processed to extract relevant features. These features can then be used to train a neural network by providing labeled examples of hydration levels associated with corresponding signal features. Through this training process, the neural network learns the complex relationship between spectral patterns and hydration status, enabling accurate, non-invasive and non-contact based hydration monitoring.

Consequently, to comprehensively analyze dehydration levels in the human body, time-frequency analysis is the most appropriate approach, as it captures the temporal variation of frequency components. This method will be employed in the following discussion.

IV. METHODOLOGY

This section outlines the development of a labeled radar dataset for noninvasive, contactless dehydration monitoring using FMCW radar systems. As there are currently no publicly available radar-based datasets labeled for dehydration assessment, we developed a custom dataset to bridge this gap. The primary challenge in building such a dataset lies in the accurate labeling of dehydration status, which typically requires clinical assessment involving medical professionals. Standard diagnostic methods, such as measuring plasma osmolality, urine osmolality, and urine specific gravity, are invasive and can only be performed in controlled clinical environments. In contrast, acquiring raw radar signals is relatively straightforward, but associating these signals with accurate hydration levels becomes impractical without medical-grade reference measurements.

This lack of reliable ground truth is a well-known barrier in radar-based machine learning applications, where large, annotated datasets are essential for training robust models. To address this issue and enable the creation of a labeled dataset with meaningful physiological variation, we adopted a novel yet practical approach: collecting radar data from individuals at various stages of voluntary fasting. Since fasting naturally induces mild to moderate dehydration, it provides a noninvasive and ethically sound framework to observe progressive hydration changes without clinical intervention. This setup allowed us to correlate the radar signals with expected hydration variation over time, offering an indirect but consistent labeling mechanism.

The following subsections detail the procedures involved in the dataset development process, including the experimental setup, data acquisition protocol, and the methods used for feature extraction and data selection.

A. Experimental Setup

The data is collected using two radar systems. The specifications of both radar modules are discussed below.

- **Inras 77 GHz FMCW-MIMO radar:** The Inras 77 GHz radar system is equipped with two transmit and sixteen receive antennas, as illustrated in Fig. 6. It transmits signals with a 2 GHz bandwidth, providing a high range resolution of 7.5 cm [30], [31]. By combining the transmit and receive antennas, a virtual antenna array can be

formed, significantly improving the angular resolution. This feature enables the system to spatially separate physiological signals, which is particularly important in multi subject environments commonly found in clinical settings. However, in this study, a simplified configuration was used for data collection: one transmit antenna and sixteen receive antennas. The radar employs sawtooth FMCW chirp modulation, which offers flexibility to adjust range resolution and Doppler sensitivity. This adaptability is crucial for dynamically tuning the system to monitor biomarkers relevant to dehydration [31], [32]. To capture reflections from the human body, a total of 256 chirp pulses are transmitted per frame, where each pulse has a duration of 512 μ s. After demodulation, the signal is sampled at 500K samples per second, resulting in 256 fast-time samples for each chirp. This collection of 256 chirps forms a single frame of receive data.

- **Analog Devices' 10GHz Phased Array radar:** This radar module operates at 10 GHz and includes one transmit and eight receive antennas, as shown in Fig. 7. Each receive antenna is a series patch design, providing a narrow beam in the elevation direction and a wide beam in the azimuth direction. The system has a bandwidth of 500 MHz, resulting in a range resolution of 30 cm. Operating at a lower frequency than 77 GHz, it experiences less atmospheric attenuation and offers deeper penetration into the human body. The phased array architecture enables electronic beam steering over $\pm 45^\circ$ in azimuth, eliminating the need for mechanical movement and allowing fast tracking of moving subjects [33]. Each FMCW pulse has a duration of 512 μ s, with $N_s = 267$ fast-time samples per chirp. This radar was used to collect a second annotated dataset following the same experimental procedure as the 77 GHz system. Using two radar systems operating at different frequencies, our aim is to broaden the evaluation of dehydration monitoring and improve the generalizability of our results. With this radar, to capture reflections from the human body, a total of 256 chirp pulses are transmitted per frame, where each pulse has a duration of 512 μ seconds. After demodulation, the signal is sampled at 500k Hz, resulting in 256 fast-time samples for each chirp. This collection of 256 chirps forms a single frame of data.

Despite hardware differences, we followed a consistent data collection protocol to ensure uniformity in processing and analysis. The resulting data set, known as the Fasting Dataset, consists of labeled radar data collected from multiple subjects.

B. Data Acquisition

For data acquisition, each subject was seated on a chair positioned 1 meter from the radar for a duration of 1–2 minutes, as illustrated in Fig. 8 and Fig. 9. In both setups, the radar module's main beam was directed toward the subject's chest, and data from a single receiving antenna and a single chirp was used to monitor dehydration.

1) **Fasting Dataset:** In the Fasting Dataset, data are collected from fasting subjects during Ramadan, when subjects



Fig. 6. Inras 77GHz FMCW radar module with 2 transmit and 16 series patch receive antennas.



Fig. 8. Experimental setup for dehydration monitoring using Inras radar Book.

fast from sunrise to sunset. During this study, we collected measurements from 16 different subjects (in 2024) on the academic campus of KAUST. All subjects fasted for approximately 12 hours, from early morning to evening, in hot weather conditions. The progressive decrease in hydration levels throughout the day provided an ideal scenario for our measurements. We took advantage of the natural decline in hydration levels from morning to night, allowing us to collect five distinct measurements at regular intervals (2 hours gap between the measurements) from each fasting subject between sunrise and sunset. The measurements are collected as follows:

- 1) **8 AM Data or DeHydL0:** The first measurement was taken when the subjects were fully hydrated and had consumed water / food before sunrise.
- 2) **10 AM Data or DeHydL1:** The second measurement was taken, representing a partially hydrated state as the fasting period began to take effect.
- 3) **12 Noon Data or DeHydL2:** The third measurement was taken at midday, indicating a mildly dehydrated state as the fasting period progressed.
- 4) **2 PM Data or DeHydL3:** The fourth measurement was

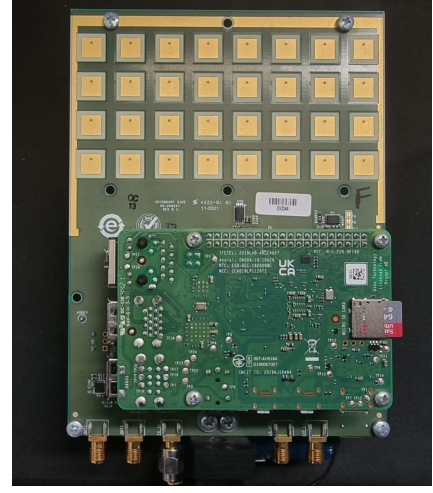


Fig. 7. Analog Devices CN0566 10 GHz phased-array radar with one transmit and 8 series patch receive antennas.



Fig. 9. Experimental setups for dehydration monitoring using CN0566 radar.

taken, reflecting a moderately dehydrated state as the effects of fasting continued.

- 5) **4 PM Data or DeHydL4:** The final measurement was taken, by which time the subjects are significantly dehydrated after nearly 12 hours of fasting.

This structured approach yields a comprehensive dataset capturing progressive changes in hydration levels throughout the day, enabling detailed analysis of dehydration patterns in fasting individuals, as illustrated in Fig. 10. As illustrated in the figure, each subject exhibits a unique pattern of weight change throughout the day, highlighting inter-individual variability—an important consideration for developing AI-based instruments capable of generalizing across a wide range of users.

C. Feature Extraction and Data Selection

The feature extraction process is designed to capture meaningful information related to the subject's hydration status. It involves two main steps: data preprocessing and WST. First, during preprocessing, we isolate the signal corresponding to

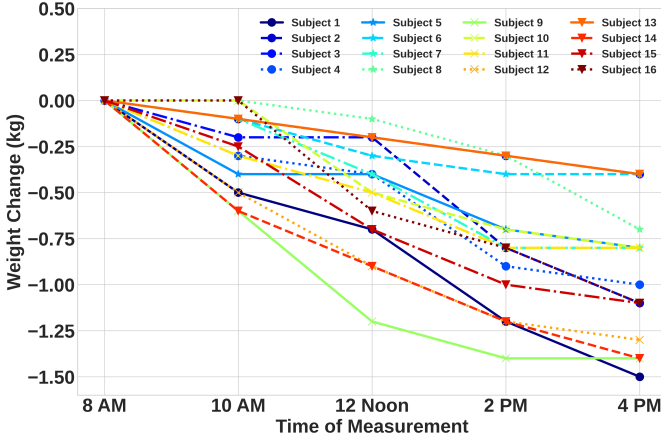


Fig. 10. Observed progressive reduction in subjects' weight from morning to evening, presented for labeling support.

the target of interest by removing noise and unwanted reflections. This helps ensure that the extracted features accurately represent the subject's physiological condition. Next, we apply the WST, a powerful time-frequency analysis method, to extract robust features. This technique captures subtle, localized changes in the radar signal that are indicative of dehydration. The detail of each step is described below:

1) *Data Pre-Processing*: The data pre-processing block involves cleaning (such as noise removal, clutter reduction, etc.) and preparing the demodulated received data to ensure it is suitable for analysis. The received demodulated signal $z(t)$ is passed through a window function (e.g., Hamming or Hanning) to reduce side lobes and improve target detection which is given by

$$x_w(t) = z(t)w(t) \quad (4)$$

where $w(t)$ represents the Hamming or Hanning window function.

The fast Fourier transform (FFT) is employed on the windowed demodulated received signal in (4) to compute the range profiles of different objects present in the scenario. In an FMCW radar system, each target can be identified by its unique beat frequency. Since the distance to the target of interest (ToI), i.e., the volunteer, is known, the corresponding beat frequency f_{bo} can be readily calculated. To isolate reflections from the immediate vicinity of the volunteer, we define a frequency window bounded by $f_{b_{min}}$ and $f_{b_{max}}$. Our goal is to retain signals within this range, specifically those satisfying $f_{b_{min}} \leq f_{bo} \leq f_{b_{max}}$ and suppress all others. This approach allows us to focus on the ToI by filtering out interference and reflections from irrelevant objects.

This is achieved by applying a second-order-section (SOS) Butterworth bandpass filter [34], [35], which is designed to provide a flat frequency response within the range $f_{b_{min}}$ to $f_{b_{max}}$ while effectively suppressing frequencies outside this band. As a result, the signal corresponding to the ToI can be effectively isolated and expressed as

$$y(t) = \text{SOS_filter}(x_w(t)). \quad (5)$$

Next, we extract the relevant features from the pre-processed data using the WST.

2) *Wavelet Scattering Transform*: The wavelet transform is a widely used time-frequency analysis method, valued for its stability and multi-scale capability, even in the presence of local deformations. While it is effective in extracting localized features from signals, it may fail to capture hierarchical or invariant representations, particularly in the presence of temporal variability. To overcome these limitations, Mallat [36] introduced the WST, an advanced technique built upon the wavelet transform framework. The wavelet scattering framework allows the extraction of low-variance features from real-valued time series data, suitable for machine learning and deep learning applications. These features are robust to input translations based on a user-defined invariance scale and remain continuous in the presence of deformations.

In essence, the Scattering Wavelet Transform (SWT) constructs a cascade of wavelet transforms and non-linear operations, similar in spirit to convolutional neural networks (CNNs), but without the need for learning or training. It is based on three fundamental operations [37]:

- 1) **Wavelet Transform (Convolution)**: The input signal is convolved with different scales of a given wavelets to extract localized frequency contents. If $\psi(t)$ is the mother wavelet, its scaled versions are defined as

$$\psi_s(t) = \frac{1}{s} \psi\left(\frac{t}{s}\right),$$

where $s = 2^j$ and $j = 0, 1, 2, \dots, J$. These scaled wavelets are created by expanding or contracting the mother wavelet to cover different frequency bands, without altering its shape. This enables the analysis of signal features at multiple resolutions.

- 2) **Non-linearity (modulus)**: The absolute value of the convolved signal is taken to ensure stability and make the representation insensitive to phase shifts.
- 3) **Low-pass filtering (averaging)**: A smoothing filter is applied to remove small-scale variations, preserving only the essential patterns at a given scale.

This sequence of operations produces a hierarchical, stable, and translation-invariant representation of the input signal. The WST provides a structured way to extract scattering coefficients of predefined orders from a signal. By leveraging multi-order scattering coefficients, WST can characterize both the coarse and fine-grained temporal dynamics of the radar return from the skin, enabling robust discrimination between hydrated and dehydrated states. Multiple order scattering coefficients can be found as

Zero-Order Scattering Coefficients: For zeroth-order scattering coefficients, the input signal is convolved with a scaling filter, $\Phi(t)$ that yield zeroth order scattering coefficients $S_0(t) = y(t) * \Phi(t)$. This operation captures the low frequencies and filter out high-frequency components of $y(t)$.

First-Order Scattering Coefficients: To capture the local variations around an arbitrary center frequency λ , the input signal is convolved with a wavelet filter $\Psi_\lambda(t)$ whose center frequency is λ . The the modulus of the resultant signal yields

the scalogram coefficients. For first order WST, the scalogram can be written as

$$U_1(t) = \{|y(t) * \Psi_{\lambda_1}(t)|\}, \quad (6)$$

where ψ_{λ_1} is wavelet filter bank for first order scattering coefficients. Each filter have different center frequency. In the next step, the 1st order scattering coefficients are determined by convolving the scalogram coefficients by the 1st order scaling filter as

$$S_1(t, \psi_{\lambda_1}) = \{|y(t) * \Psi_{\lambda_1}(t)| * \Phi(t)\} \quad (7)$$

The first order scattering coefficients, S_1 , yields the information about the low-frequency components of U_1 .

Second-Order Scattering Coefficients: To recover the high-frequency information lost in S_1 , U_1 is convolved with wavelet filter bank $\psi_2(t)$, which has multiple wavelets, to obtain second order scalogram coefficients as

$$U_2(t) = \{|U_1(t) * \Psi_{\lambda_2}(t)|\} \quad (8)$$

and correspondingly second-order scattering coefficients are obtained as

$$S_2(t, \psi_{\lambda_1}, \psi_{\lambda_2}) = \{U_2(t) * \Phi(t)\}.$$

By iterating the above process, the wavelet scattering coefficients can be defined for order m as

$$S_m(t, \psi_{\lambda_1}, \psi_{\lambda_2}, \dots, \psi_{\lambda_m}) = \{U_m(t) * \Phi(t)\} \quad (9)$$

where $U_m(t) = \left| \dots |y(t) * \Psi_{\lambda_1}(t)| * \Psi_{\lambda_2}(t)| \dots \Psi_{\lambda_m}(t) \right|$. Each subsequent order captures increasingly complex structures such as modulations of modulations, while maintaining stability and translation invariance through the use of a low-pass filter at the final stage of each order.

After extracting features using the WST, the next step is to feed these features into multiple classifiers to identify the dehydration status.

V. DEHYDRATION CLASSIFICATION

In this section, we detail the final component of our proposed methodology: classification. We train and evaluate a comprehensive range of machine learning and deep learning models to accurately classify dehydration levels. The machine learning methods we implemented include K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Naive Bayes, and Neural Networks (NN). Each method is selected for its unique ability to capture diverse patterns within the radar data, essential for a precise classification of dehydration. By rigorously training and testing these models, we aim to determine the most effective algorithm for reliable and efficient noninvasive dehydration monitoring.

A. K-Nearest Neighbors

We advocate for KNN to classify WST features from the Inras 77 GHz and CN0566 10 GHz radars due to their synergy with the nonlinear, high-dimensional nature of WST outputs and the localized physiological patterns related to dehydration. WST generates translation-invariant hierarchical representations of radar signals, preserving subtle biomarkers (e.g., cardiopulmonary motion, tissue dielectric shifts) while suppressing noise. These features exhibit complex distributions, particularly when derived from multiband radars: the 77 GHz system resolves subcentimeter thoracic movements, while the CN0566 phased array captures angular-dependent postural changes. The nonparametric framework of KNN avoids assumptions about data linearity or global separability, instead utilizing local similarity metrics (e.g., Mahalanobis distance) to identify hydration-related clusters. This aligns with the gradual hemodynamic effects of dehydration. The minimal training requirements of KNN mitigate the risks inherent in overfitting smaller clinical data sets, as it lacks internal parameters beyond the k selection and distance metrics. The inherent dimensionality reduction of WST further counteracts the 'curse of dimensionality', allowing KNN to generalize effectively without deep architectures. Additionally, KNN supports real-time inference via optimized data structures (e.g., k -d trees), which is critical for embedded deployment with radar systems. Its interpretability, classifying based on proximity to labeled neighbors, enables clinical validation of dehydration thresholds (for example, correlating characteristics with 2-3% body mass loss). Finally, KNN permits radar-specific model tuning (e.g., distinct k values for 77 GHz vs. 10 GHz features) while enabling late fusion for multisensor integration, ensuring scalability.

B. Support Vector Machine

WST generates translation-invariant, deformation-stable features by cascading wavelet convolutions and modulus operations. These coefficients form high-dimensional manifolds that encode hydration biomarkers: mmWave reflections (77 GHz) correlate with superficial tissue permittivity, while C-band signals (10 GHz) probe deeper dielectric properties. SVMs maximize interclass margins in such spaces, prioritizing structural risk minimization. This is critical for limited biomedical datasets, ensuring robust generalization even with sparsely labeled examples. The Inras mmWave reflections depend non-linearly on the epidermal water content, while the CN0566 data capture nonstationary scattering in deeper tissues. SVMs address this via kernel methods, such as the Radial Basis Function (RBF), which implicitly maps features into separable spaces. This avoids manual feature engineering when multimodal data are fused, enabling joint modeling of high-frequency (77 GHz) and low-frequency (10 GHz) biomarkers. SVMs outperform deep learning in small-data regimes, common in clinical studies. Kernelized SVMs leverage the kernel trick for tractable computation, enabling iterative hyperparameter tuning (e.g., regularization, kernel bandwidth) across dual radar systems. Additionally, SVM decision boundaries offer interpretability: Support vec-

tors highlight discriminative WST nodes, such as those tied to dermal thickness (77 GHz) or blood volume (10 GHz), validating physiological relevance.

C. Neural Networks

NNs address critical challenges posed by physiological radar data, including nonlinear relationships, multiscale dependencies, and multimodal fusion, while enabling robust real-time deployment. Dehydration alters subtle physiological signals (e.g., micro-Doppler signatures, tissue dielectric properties) in non-linear ways due to changes in blood viscosity and reduced skin elasticity. WST compresses raw radar data into stable, invariant features, but these span high-dimensional scattering paths. NNs, with stacked nonlinear activation layers (e.g., ReLU, tanh), autonomously learn hierarchical representations, disentangling complex dehydration markers from noise or motion artifacts. The inherent translation invariance of WST preserves discriminative patterns on all scales: critical for radar data, where dehydration manifests differently in superficial (77 GHz) versus deep tissue (10 GHz) signals. NNs dynamically prioritize scale-specific features: shallow layers may amplify high-frequency skin tremors, while deeper layers correlate slow hemodynamic shifts. The feedforward architecture of NNs and the compatibility with edge computing frameworks (e.g. TensorFlow Lite) ensure low-latency inference, crucial for continuous hydration monitoring. Unlike memory-intensive kernel methods, optimized NNs are easily deployed on embedded GPUs or FPGAs, aligning with the constraints of portable healthcare devices.

D. Naive Bayes

Naive Bayes's (NB) probabilistic framework, based on conditional independence assumptions, enables rapid training and inference - critical for real-time dehydration alerts. WST generates high-dimensional matrices from multilayered radar returns (e.g. range-Doppler maps), but NB minimizes computational overhead without iterative optimization. This aligns with the latency and power constraints of embedded radar systems. Empirical studies in Doppler classification show NB achieves $> 95\%$ accuracy with small datasets, a key advantage given the limited labeled hydration data. Radar biosignals are corrupted by motion artifacts, multipath interference, and RF noise. Although WST's averaging layers suppress high-frequency noise, redundant coefficients from overlapping wavelets persist. NB's independence assumption inherently reduces overfitting by treating correlated features as conditionally independent, particularly beneficial for the Inras 77 GHz radar's cluttered mmWave spectrum. Probabilistic output also enables adaptive confidence thresholds, distinguishing dehydration-induced permittivity changes from transient sweat artifacts.

VI. EXPERIMENTS

This section describes the experimental protocols employed in the integration and deployment of the Inras 77 GHz radar system alongside the CN0566 Phased Array 10 GHz radar

modules. Subsequently, we present a comprehensive comparative analysis that assesses the performance metrics of both radar systems when subjected to empirically defined real-world parameters and constraints.

A. Performance of the machine learning models

1) *Overall performance of Inras 77GHz radar::* Data collection is conducted using the Inras 77 GHz radar, which produces a data matrix of size $256 \times 256 \times 16$. For our analysis, we selected the fast-time samples of a specific chirp and receiving antenna, resulting in a 1×256 data vector. The initial step involves pre-processing the received demodulated data to remove noise and interference. This is achieved by passing the signal through a second-order section (SOS) Butterworth bandpass filter, isolating the target at 1.2 meters in the Fasting dataset. This filtered signal contains the relevant information related to the subject, or target of interest. Next, the filtered signal is fed to a wavelet scattering transform to extract meaningful features related to the target of interest. Finally, these extracted features are input into various machine learning classifiers, including Naive Bayes, KNN, SVM, and NN, to determine the dehydration status at 5 levels for the Fasting dataset.

For the Fasting datasets, we utilized Gabor (analytic Morlet) wavelets to perform wavelet decomposition, with a Gaussian function serving as the corresponding low-pass filter. The invariance scale is set to the default i.e., half the signal length (128 samples). The constructed wavelet scattering network comprised two filter banks. Specifically, we use Q factor [8 1] for the Fasting Dataset. Although various settings for the invariance scale and wavelet octave resolution are tested, this configuration preserved the signal information best for classification.

For the Fasting dataset, the wavelet scattering network's output forms a matrix of size $31 \times 8 \times 10000$. Each slice of the tensor represents the scattering coefficients containing the dehydration information of the signal. These coefficients are also down-sampled based on the low-pass filter's bandwidth, producing 8 time windows for each of the 31 scattering paths. To align with the classifiers data structure requirements and for better results, we reshaped this matrix into a 10000×248 , where each column corresponds to a scattering path and a time window. The 248 columns result from the 8 time windows for each of the 10000 signals in the Fasting dataset. We employed a 5-fold cross-validation method for both datasets, splitting 90% of the data for training and reserving the remaining 10% for testing. Our proposed methodology was evaluated using Naive Bayes, SVM, KNN, and NN. All classifiers except Naive Bayes performed effectively in distinguishing dehydration levels, achieving high accuracy for the Fasting dataset (5-level classification). Notably, even basic classification methods demonstrated strong performance on the 5-level classification of the Fasting Dataset.

The performance of four different classifiers NN, KNN, Naive Bayes, and SVM are evaluated using confusion matrix (shown in Fig. 11, 12, 13, 14) and standard classification metrics, including precision, recall, F1-score, accuracy, and misclassification rate as shown in Table II, III, IV, V.

Class Name	Precision	Recall	F1 score
DeHydL0	0.7236	0.7200	0.7218
DeHydL1	0.6402	0.7650	0.6970
DeHydL2	0.7000	0.6300	0.6632
DeHydL3	0.8652	0.6100	0.7155
DeHydL4	0.6515	0.7850	0.7120
Accuracy	0.7020		
Macro-F1	0.7019		

TABLE II

77GHZ RADAR PERFORMANCE METRICS FOR BAYES

Class Name	Precision	Recall	F1 score
DeHydL0	0.9849	0.9800	0.9825
DeHydL1	0.9846	0.9600	0.9722
DeHydL2	0.9431	0.9950	0.9684
DeHydL3	0.9949	0.9750	0.9848
DeHydL4	0.9849	0.9800	0.9825
Accuracy	0.9780		
Macro-F1	0.9781		

TABLE IV

77GHZ RADAR PERFORMANCE METRICS FOR KNN

Class Name	Precision	Recall	F1 Score
DeHydL0	1.0000	0.9300	0.9637
DeHydL1	0.8889	1.0000	0.9412
DeHydL2	0.9898	0.9750	0.9824
DeHydL3	0.9948	0.9600	0.9771
DeHydL4	0.9899	0.9850	0.9875
Accuracy	0.9700		
Macro-F1	0.9704		

TABLE III

77GHZ RADAR PERFORMANCE METRICS FOR SVM

Class Name	Precision	Recall	F1 Score
DeHydL0	0.9608	0.9800	0.9703
DeHydL1	0.9561	0.9800	0.9679
DeHydL2	0.9898	0.9750	0.9824
DeHydL3	0.9849	0.9800	0.9825
DeHydL4	0.9897	0.9650	0.9772
Accuracy	0.9760		
Macro-F1	0.9760		

TABLE V

77GHZ RADAR PERFORMANCE METRICS FOR NN

The Naive Bayes classifier exhibits an accuracy of 70.2%, with a macro-F1 score of 0.7019. It demonstrates moderate performance, with Class 3 (DeHydL3) achieving the highest precision (0.8652) and recall (0.6100). The overall F1 score for this class is 0.7155. However, other classes, such as Class 1 (DeHydL1) and Class 2 (DeHydL2), show relatively lower precision and recall scores, especially in comparison to the other classifiers. Class 1, for instance, has a precision of 0.6402 and recall of 0.7650, indicating room for improvement in the classifier's ability to correctly identify this class.

The SVM classifier significantly outperforms Naive Bayes with an impressive accuracy of 97.0% and a macro-F1 score of 0.9704. The precision and recall values across all classes are consistently high, particularly for Class 1 (DeHydL1), which achieved perfect recall (1.0000) and high precision (0.8889), resulting in an F1 score of 0.9412. Class 0 (DeHydL0) also performs exceptionally well with a precision of 1.0000 and recall of 0.9300, yielding an F1 score of 0.9637. Overall, the SVM classifier exhibits robust classification capabilities, showing excellent generalization across all classes (shown in Fig. 12).

The KNN classifier also performs admirably with an accuracy of 97.8% and a macro-F1 score of 0.9781. Precision and recall values for all classes are very strong, with Class 2 (DeHydL2) achieving near-perfect precision (0.9431) and recall (0.9950), leading to an outstanding F1 score of 0.9684. Similar to SVM, KNN shows consistent performance across the different classes, with Class 3 (DeHydL3) achieving an F1 score of 0.9848, reflecting its high precision (0.9949) and recall (0.9750). KNN, therefore, presents a competitive alternative to SVM with only marginal differences in performance.

The NN classifier achieves a near-identical performance to KNN, with an accuracy of 97.6% and a macro-F1 score of 0.9760. The precision and recall for most classes are close to the highest observed, with Class 0 (DeHydL0) yielding an F1 score of 0.9703, which is slightly lower than SVM and

KNN. Class 1 (DeHydL1) exhibits a strong recall of 0.9800, resulting in a precision of 0.9561 and an F1 score of 0.9679. NN provides high overall performance but does not outperform the other models in any particular area.

Overall, integrating WST features significantly improved classification performance. The SVM, KNN, and NN classifiers achieved high accuracy and reliable classification results, making them well-suited for dehydration level detection based on radar data. The poor performance of Naive Bayes suggests that it is less effective for this application due to its underlying assumptions about feature independence. The results demonstrate the importance of selecting appropriate machine learning models and feature extraction techniques for robust non-invasive dehydration monitoring.

2) *Overall performance of CN0566 Phased Array radar:* In our study, we evaluated the performance of Naive Bayes, SVM, KNN, and NN classifiers trained on data collected via the CN0566 10GHz Phased Array radar to classify dehydration levels into five distinct classes during Ramadan. The training set comprised 9000 samples and for testing 1000 samples, from 20 participants, collected at five intervals (8 AM, 10 AM, 12 PM, 2 PM, 4 PM) to account for temporal variations in dehydration under fasting conditions.

In our study, we evaluated the performance of a Naive Bayes classifier for non-contact dehydration monitoring using the CN0566 Phased Array FMCW radar and Wavelet Scattering Transform (WST). The model achieved a general precision of 59% (misclassification rate: 41%), as shown in Table VI, with macro- and weighted-F1 scores of 0.59, indicating moderate discriminative capacity in five levels of dehydration (DeHydL0–DeHydL4). Class-specific analysis revealed pronounced performance disparities: DeHydL3 and DeHydL4 exhibited higher precision (0.73, 0.70) and recall (0.73, 0.70), yielding robust F1 scores (0.73, 0.70), while DeHydL1 and DeHydL2 performed at chance levels (F1: 0.50). DeHydL0 showed intermediate results (F1: 0.52), suggesting partial

DeHydL0	144	15	6	6	29
DeHydL1	13	153	17	7	10
DeHydL2	19	24	126	5	26
DeHydL3	11	32	16	122	19
DeHydL4	12	15	15	1	157
	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4

Predicted Class

Fig. 11. Confusion matrix for Naive Bayes (77GHz radar)

DeHydL0	186	11	1		2
DeHydL1		200			
DeHydL2		4	195	1	
DeHydL3		8		192	
DeHydL4		2	1		197
	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4

Predicted Class

Fig. 12. Confusion matrix for SVM (77GHz radar)

DeHydL0	196		3		1
DeHydL1	2	192	4	1	1
DeHydL2	1		199		
DeHydL3		2	2	195	1
DeHydL4		1	3		196
	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4

Predicted Class

Fig. 13. Confusion matrix for KNN (77GHz radar)

DeHydL0	196	2	1	1	
DeHydL1	1	196	1	1	1
DeHydL2	1	4	195		
DeHydL3	1	2		196	1
DeHydL4	5	1		1	193
	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4

Predicted Class

Fig. 14. Confusion matrix for NN (77GHz radar)

separability. The confusion matrix, as shown in Figure 15 highlighted systematic misclassifications: DeHydL0 was frequently confused with DeHydL1 and DeHydL2 (35 and 29 instances, respectively), while DeHydL1 and DeHydL2 exhibited cross-classification (e.g., DeHydL1 misclassified as DeHydL0: 30, DeHydL2: 34). This suggests overlapping WST-derived features in early stages of dehydration, likely due to subtle physiological variations in radar-reflected signals. In contrast, DeHydL3 and DeHydL4 demonstrated higher specificity, with 146 and 14 true positives, respectively, underscoring the efficacy of the model in detecting advanced dehydration. The parity between macro- and weighted-F1 (0.59) implies class imbalance (e.g., DeHydL0: 182 samples, DeHydL4: 74) did not disproportionately affect metrics, though the Naive Bayes assumption of feature independence may limit performance given correlated WST coefficients. These results validate the feasibility of radar-based dehydration monitoring, but emphasize the need for advanced models (e.g. ensemble methods) to disentangle low-level dehydration signatures while retaining robustness in critical cases.

In our investigation of the SVM model for non-contact dehydration monitoring using the CN0566 Phased Array FMCW radar and WST, we observed a marked improvement in performance compared to prior Naive Bayes results. The SVM achieved an overall precision of 93.5% (misclassification rate: 6.5%, Table VII), with uniformly high macro and weighted F1 scores of 0.935, demonstrating robust class discrimination across all levels of dehydration (DeHydL0–L4).

DeHydL0	104	35	29	12	2
DeHydL1	30	100	34	22	14
DeHydL2	14	54	100	9	23
DeHydL3	40	9	2	146	3
DeHydL4	12	2	35	11	14
	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4

Predicted Class

Fig. 15. Confusion matrix for Naive Bayes (10GHz radar)

DeHydL0	191	3	1	3	2
DeHydL1	3	188	4	3	2
DeHydL2	3	4	180	11	2
DeHydL3	1		15	183	1
DeHydL4	2	5			193
	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4

Predicted Class

Fig. 16. Confusion matrix for SVM (10GHz radar)

Class-specific metrics revealed exceptional precision and recall values (> 0.90 for all classes), with DeHydL4 achieving

Class Name	Precision	Recall	F1 Score
DeHydL0	0.52	0.52	0.52
DeHydL1	0.50	0.50	0.50
DeHydL2	0.50	0.50	0.50
DeHydL3	0.73	0.73	0.73
DeHydL4	0.70	0.70	0.70
Accuracy	0.59		
Macro-F1	0.59		

TABLE VI
PERFORMANCE METRICS FOR NAIVE BAYES

near-perfect scores (precision: 0.965, recall: 0.965, F1: 0.965), underscoring the model's reliability in detecting severe dehydration. The confusion matrix (Figure 16) exhibited minimal off-diagonal misclassifications, with DeHydL0–L2 showing fewer than 5 errors in most cases (e.g., DeHydL0: 191 true positives, 3 misclassified as L1), and DeHydL3–L4 showing high specificity (183 and 193 true positives, respectively). In particular, misclassifications were restricted to adjacent classes (e.g., DeHydL2 confused with L3: 11 instances), likely due to overlapping physiological signatures in the moderate stages of dehydration. The parity between macro- and weighted-F1 scores suggests a balanced class representation or the ability of the SVM to mitigate bias through margin maximization, even with potential feature correlations in WST-derived coefficients. We attribute this performance to the SVM's kernel-driven separation of nonlinear feature spaces, possibly using an RBF kernel to handle the complex relationships in radar-reflected signals processed via WST. The results validate the efficacy of combining high-resolution phased array radar data with WST's invariance properties to extract discriminative hydration-related features, such as tissue permittivity changes or micro-motion patterns. This approach significantly advances noninvasive dehydration monitoring, particularly in critical cases (L3–L4), while maintaining generalizability throughout the early stages.

In our evaluation of the KNN classifier for noncontact dehydration monitoring using the CN0566 Phased Array FMCW radar and Wavelet Scattering Transform (WST), we achieved an accuracy of 95.5% (misclassification rate: 4.5%, Table VIII), with macro- and weighted-F1 scores of 0.955, reflecting strong class-agnostic performance across all levels of dehydration (DeHydL0–L4). The model demonstrated near-perfect precision and recall for DeHydL0 (F1: 0.98), DeHydL3 (F1: 0.99) and DeHydL4 (F1: 0.975), indicating exceptional discriminative capability in detecting baseline, severe, and critical dehydration states. DeHydL1 and L2 exhibited marginally lower but robust metrics (F1: 0.91, 0.92), suggesting moderate interclass ambiguity in early- to mid-stage dehydration. The confusion matrix, as shown in Figure 17, revealed localized misclassifications: DeHydL1 was frequently confused with L2 (15 instances) and L0 (9 instances), while DeHydL2 showed cross-classification with L1 (10 instances) and L0 (3 instances). These errors likely result from overlapping WST-derived feature distributions in adjacent hydration stages, where subtle physiological variations, such as tissue permittivity or micro-Doppler signatures, may not be fully separable under the distance-based voting mechanism of KNN. In con-

Class Name	Precision	Recall	F1 score
DeHydL0	0.955	0.955	0.955
DeHydL1	0.940	0.940	0.940
DeHydL2	0.900	0.900	0.900
DeHydL3	0.915	0.915	0.915
DeHydL4	0.965	0.965	0.965
Accuracy	0.935		
Macro-F1	0.935		

TABLE VII
PERFORMANCE METRICS FOR SVM

True Class \ Predicted Class	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4
DeHydL0	196	3	1		
DeHydL1	9	182	15	1	2
DeHydL2	3	10	184	1	2
DeHydL3	1			198	1
DeHydL4		5			195

Fig. 17. Confusion matrix for KNN (10GHz radar)

True Class \ Predicted Class	DeHydL0	DeHydL1	DeHydL2	DeHydL3	DeHydL4
DeHydL0	197	1	1	1	
DeHydL1	1	198			1
DeHydL2			194	3	3
DeHydL3	2		5	193	
DeHydL4		1		3	196

Fig. 18. Confusion matrix for NN (10GHz radar)

trast, DeHydL3 and L4 achieved near-ideal specificity (198 and 195 true positives, respectively), underscoring the model's ability to resolve advanced dehydration markers, which exhibit more pronounced deviations of radar-reflected signals. The parity between macro- and weighted-F1 scores implies balanced class representation or effective handling of minor imbalances (e.g., DeHydL3: 200 samples, L4: 200), though the sensitivity of the KNN to feature scaling and local noise may explain residual misclassifications in transitional stages. The results validate the synergy between the translation-invariant feature extraction of WST and the nonparametric design of KNN, which together capture hierarchical patterns related to hydration in phased array radar data.

In our evaluation of the NN model for non-contact dehydration monitoring using the CN0566 Phased Array FMCW radar and Wavelet Scattering Transform (WST), we achieved exceptional performance, with an accuracy of 97.8% (misclassification rate: 2.2%, Table IX) and near-perfect macro- and weighted F1 scores of 0.978. These metrics underscore the robust ability of the model to discriminate between all stages of

Class Name	Precision	Recall	F1 Score
DeHydL0	0.980	0.980	0.980
DeHydL1	0.910	0.910	0.910
DeHydL2	0.920	0.920	0.920
DeHydL3	0.990	0.990	0.990
DeHydL4	0.975	0.975	0.975
Accuracy	0.955		
Macro-F1	0.955		

TABLE VIII
PERFORMANCE METRICS FOR KNN

dehydration (DeHydL0-L4), using the high-resolution micro-Doppler and permittivity-derived features of the radar. Class-specific precision and recall exceeded 0.96 for all classes, with DeHydL0 (F1: 0.985), DeHydL1 (F1: 0.990), and DeHydL4 (F1: 0.980) demonstrating near-flawless classification. The confusion matrix, as shown in Figure 18, revealed minimal off-diagonal errors, dominated by localized ambiguities. DeHydL3 exhibited five misclassifications as L2, while DeHydL2 showed 3 instances confused with L3, probably due to overlapping physiological signatures (e.g. tissue elasticity or subtle fluid loss patterns) in mid to advanced stages of dehydration. In particular, DeHydL0-L1 achieved near-ideal specificity (197 and 198 true positives, respectively), while DeHydL4 maintained 193 correct identifications, highlighting the precision of the model in detecting baseline and critical hydration states. The parity between macro- and weighted-F1 scores suggests balanced class distributions or the ability of NN to mitigate bias through hierarchical feature learning, even with potential nonlinear correlations in WST-derived coefficients. We attribute this performance to the NN's capacity to model complex decision boundaries in radar-reflected signals, which encode hydration-related variations such as skin-layer permittivity changes or capillary flow dynamics. The translation invariance of the WST probably enhanced the stability of the characteristics, allowing the NN to extract discriminative patterns in varying signal-to-noise ratios. Residual errors in DeHydL2-L3 may arise from transitional physiological states where radar signatures exhibit spectral overlap, challenging even deep architectures. These results validate the synergy between the spatial resolution of the phased array radar and the nonlinear modeling of NN for dehydration monitoring, outperforming previous implementations of SVM and KNN.

3) Impact of different Scattering Filter Bank for 77GHz::

To assess the impact of WST filter banks on classification accuracy, we analyzed the performance of different classifiers using various WST configurations, as summarized in Table X. The results indicate that increasing the quality factor Q of the scattering filter bank significantly improves classification accuracy across all methods. For the lowest filter bank setting, WST_Q21, classification accuracies remained relatively modest, with Naive Bayes achieving 42.3%, while more advanced classifiers such as SVM, KNN, and NN exhibited 81.2%, 79%, and 90.9%, respectively. However, as the quality factor increased to WST_Q41, a substantial improvement was observed, with accuracies reaching 94.9% (SVM), 93.5% (KNN), and 96.6% (NN). This trend continued with WST_Q81, where the highest classification performance

Class Name	Precision	Recall	F1 score
DeHydL0	0.985	0.985	0.985
DeHydL1	0.990	0.990	0.990
DeHydL2	0.970	0.970	0.970
DeHydL3	0.965	0.965	0.965
DeHydL4	0.980	0.980	0.980
Accuracy	0.978		
Macro-F1	0.978		

TABLE IX
PERFORMANCE METRICS FOR NN

was achieved, particularly for KNN (97.8%), followed closely by NN (97.6%) and SVM (97%).

WST Filter bank	Naive Bayes	SVM	KNN	NN
WST_Q21	42.3	81.2	79	90.9
WST_Q41	56.6	94.9	93.5	96.6
WST_Q81	70.2	97	97.8	97.6

TABLE X
ACCURACY OF DIFFERENT CLASSIFIERS USING VARIOUS WST CONFIGURATIONS FOR 77 GHZ RADAR

These results clearly demonstrate that increasing the filter bank quality factor enhances the discriminative power of the extracted features, leading to improved classification performance. Notably, the Naive Bayes classifier, which relies on strong independence assumptions, showed a more significant improvement compared to its lower- Q counterparts but still lagged behind more sophisticated models. Overall, the integration of WST not only facilitates effective dimensionality reduction but also enhances classification accuracy by providing more informative and structured feature representations. The observed improvements further highlight the robustness of WST in radar-based classification tasks, making it a crucial component of our feature extraction pipeline.

4) Impact of Different Scattering Filter Bank for 10GHz::

The results presented in Table XI illustrate the precision of various classifiers: Naive Bayes, SVM, KNN, and NN utilizing different configurations of the WST filter bank for a 10 GHz radar system. We observed that as we increased the complexity of the WST filter bank from Q21 to Q81, there was a notable enhancement in the classification accuracy in all classifiers. Specifically, the SVM classifier achieved the highest accuracy of 93.5% with the Q81 configuration, indicating its robustness in handling the extraction of features provided by the more intricate wavelet structure. Furthermore, the KNN and NN classifiers also demonstrated significant improvements, with accuracy of 95.5% and 97.8%, respectively, when using the Q81 configuration. This suggests that the increased number of wavelet filters enhances the model's ability to capture intricate patterns in the data, thereby improving classification performance. In contrast, the Naive Bayes classifier exhibited the lowest accuracy in all configurations, with a maximum of 59% for Q81, highlighting its limitations in scenarios where complex feature interactions are present. In general, our findings underscore the importance of selecting appropriate WST configurations to optimize classifier performance in radar applications. The progressive increase in accuracy with more sophisticated filter banks indicates that leveraging advanced

wavelet techniques can significantly enhance the interpretability and effectiveness of machine learning models in processing radar data.

WST Filter bank	Naive Bayes	SVM	KNN	NN
WST_Q21	50.1	88.9	90.2	91.4
WST_Q41	53.2	92.8	94.2	96.2
WST_Q81	59	93.5	95.5	97.8

TABLE XI
ACCURACY OF DIFFERENT CLASSIFIERS USING VARIOUS WST CONFIGURATIONS FOR 10 GHZ RADAR

5) *Comparison of machine learning models for 77GHz and 10GHz radar:* The overall performance trends were consistent across both radar frequencies. Naive Bayes exhibited the weakest results at both 10 GHz (59% accuracy) and 77 GHz (70.2% accuracy), struggling due to its assumption of feature independence, which poorly accommodated the correlated radar features. The SVM classifier performed robustly at both frequencies, with accuracy of 93.5% at 10 GHz and 97.0% at 77 GHz, showing its strong capability in handling nonlinear feature spaces. The KNN model also performed well, achieving 95.5% accuracy at 10 GHz and 97.8% at 77 GHz, with minimal differences between the two frequencies. The NN classifier outperformed all other models at both frequencies, with 97.8% accuracy at 10 GHz and 97.6% at 77 GHz. Although its performance was slightly lower at 77 GHz, it still demonstrated high precision and recall across all classes, showcasing its ability to model complex physiological features effectively. In summary, the SVM, KNN, and NN classifiers exhibited similar performance across both radar frequencies, with NN showing the highest accuracy overall. The 77 GHz radar provided finer resolution, but the performance differences between the two frequencies were minimal, with all models showing strong results for dehydration monitoring. Future work should focus on optimizing these models for real-time, embedded systems, particularly NN, which shows the most promise for clinical applications.

6) *Comparison of machine learning models:* In our comparative analysis of machine learning models for non-contact dehydration monitoring using the CN0566 Phased Array FMCW radar and Wavelet Scattering Transform (WST), distinct performance hierarchies emerged. The Naive Bayes (NB) classifier exhibited the weakest results, with an accuracy of 59% (F1: 0.59), due to its inherent assumption of feature independence, which poorly accommodated the correlated radar features derived from WST. Systematic misclassifications between early stages of dehydration (DeHydL0–L2) underscored its inability to resolve subtle physiological variations, such as minor permittivity shifts in skin layers. In contrast, the SVM achieved robust performance (93.5% accuracy, F1: 0.935), leveraging kernel-driven separation to handle nonlinear feature spaces, likely via an RBF kernel. Its confusion matrix showed minimal errors, mainly between adjacent classes (for example, DeHydL2–L3), attributed to transitional physiological signatures. The KNN model demonstrated intermediate efficacy (95.5% precision, F1: 0.955), with localized ambiguities in DeHydL1–L2 due to overlapping feature distributions in mid-stage dehydration, where distance-based voting struggled to

discriminate nuanced radar-reflected micro-Doppler patterns. Finally, the NN outperformed all models (97.8% accuracy, F1: 0.978), capitalizing on its capacity to model hierarchical, nonlinear relationships in WST-processed radar data. Near-perfect precision and recall across all classes, with residual errors confined to DeHydL2–L3 transitions, highlighted its superiority in disentangling complex hydration biomarkers, such as tissue elasticity gradients and capillary flow dynamics. Although SVM and KNN offered practical trade-offs between interpretability and computational efficiency, the depth and adaptability of the NN proved critical for the monitoring of high accuracy and real-time dehydration. These findings advocate for the deployment of NN architectures in clinical settings, although future work should optimize latency for embedded systems and investigate hybrid models (e.g., CNN-SVM) to mitigate transitional-stage ambiguities.

7) *Comparison with related works:* To give the reader a clear picture, Table XII compares our non-contact and non-invasive dehydration monitoring system with current state-of-the-art techniques. This comparison considers several important aspects: the sensing modality, the accuracy achieved, the number of classes it can distinguish, the strategy used for labeling data, the number of subjects involved in testing, and the evaluation methodology. We believe that our work has some key advantages, including the following:

- Unlike previous studies that used contact-based sensors (e.g., PPG, EDA) or camera-based vision analysis, *we are the first to use 10 GHz and 77 GHz radar data for non-contact monitoring.*
- Although prior research classified dehydration into only 2, 3, or 4 levels, *our approach uniquely estimates dehydration on a finer scale of 1-to-5.*
- We also introduce a novel publicly available labeled dehydration dataset (available from the corresponding author upon reasonable request), a first for the research community to use for future experiments.
- The proposed contactless method is particularly promising for situations requiring minimal physical interaction (e.g., during pandemics) and enables unobtrusive monitoring of vulnerable individuals like the elderly or individuals at high risk due to exertion like athletes.

In addition, this method could be incorporated into the health vertical of the upcoming 6G cellular standard, contributing to its integrated sensing and communication (ISAC) capabilities.

VII. CONCLUSION AND DISCUSSION

A. Conclusion

This paper presents a novel noninvasive approach for dehydration monitoring using FMCW radar operating at 10 GHz and 77 GHz. By integrating a preprocessing technique to isolate the target of interest and employing WST for feature extraction, we significantly improved the accuracy of dehydration classification. Our methodology, validated with custom "Fasting Datasets" under controlled conditions, demonstrates the feasibility of using radar-based systems for non-contact dehydration monitoring. The comparative analysis of machine

TABLE XII
COMPARISON OF THE PROPOSED RADAR-BASED METHOD FOR NON-INVASIVE DEHYDRATION MONITORING WITH THE STATE-OF-THE-ART METHODS.

Work	Method	Modality	Acc. (%)	No. of Classes	Labeling Strategy	No. of Subjects	Evaluation Methodology
[7]	Non-invasive	EDA	97.83	3	Fasting-based time-dependent	16	Segment-level random split (subject-dependent)
[10]	Non-invasive	EDA	84.5	3	Fasting-based time-dependent	5	Segment-level random split (subject-dependent)
[11]	Non-invasive	EDA	91.53	2	Fasting-based time-dependent	5	K-fold cross-validation (subject-dependent)
[12]	Non-invasive	EDA	87.78	2	Fasting-based time-dependent	5	K-fold cross-validation (subject-dependent)
[38]	Non-invasive	Facial images	76.1	3	Fasting-based time-dependent	70	random split + K-fold cross-validation (subject-independent)
[39]	Non-invasive	Acoustic	91.5	2	activity-based labeling (manual annotation)	8	K-fold cross-validation (subject-independent)
[8]	Non-invasive	PPG	67.91	2	Clinical diagnosis	42	K-fold cross-validation (subject-dependent)
[17]	Non-contact	RF	96.15	2	Fasting-based time-dependent	5	K-fold cross-validation (subject-dependent)
[14]	Non-invasive	video-PPG	99.37	4	Fasting-based time-dependent	50	Segment-level random split (subject-dependent)
[13]	Non-invasive	1-lead ECG	97.41	4	Fasting-based time-dependent	55	Segment-level random split (subject-dependent)
[15]	Non-invasive	capacitance	75	4	Fasting-based time-dependent	35	Segment-level random split (subject-dependent)
This work	Non-contact	10 GHz radar	97.8	5	Fasting-based time-dependent	16	K-fold cross-validation (subject-dependent)
This work	Non-contact	77 GHz radar	97.6	5	Fasting-based time-dependent	16	Segment-level random split (subject-dependent)

learning classifiers, including Naive Bayes, SVM, KNN, and NN, revealed robust performance across both radar frequencies. While Naive Bayes struggled due to its assumptions about feature independence, SVM, KNN, and NN exhibited strong classification capabilities, with NN achieving the highest accuracy overall (97.8% at 10 GHz and 97.6% at 77 GHz). The 77 GHz radar, with its finer resolution, demonstrated superior performance, but the differences between the two frequencies were minimal, emphasizing the effectiveness of our approach at both radar frequencies.

The results highlight the importance of feature extraction techniques, such as WST, in capturing the complex, nonlinear characteristics of radar data for dehydration classification. This study demonstrates that the combination of high-frequency radar, advanced feature extraction methods, and machine learning classifiers holds great potential for non-invasive, non-contact monitoring of dehydration levels. Future work should focus on optimizing these models for embedded systems to enhance their applicability in clinical and remote health monitoring applications. Ultimately, our findings suggest that radar-based dehydration monitoring, particularly with neural network classifiers, could serve as a valuable tool in healthcare settings, offering a promising non-invasive method for continuous monitoring of hydration status. Further research and development in this area could lead to impactful solutions for early detection and management of dehydration, particularly in resource-limited or remote environments.

B. Limitations and future works

In this section, we critically analyze the technical and practical limitations of our FMCW radar-based dehydration monitoring framework, which integrates Wavelet Scattering Transform (WST) and machine learning. Although the system demonstrates promising accuracy in controlled settings, challenges persist in environmental robustness, computational efficiency, and clinical generalizability. We also outline a systematic roadmap for future research that addresses adaptive noise suppression, optimized WST architectures, and large-scale multimodal validation.

1) *Limitations:* These are some of the limitations:

- FMCW radars, while effective in capturing physiological signals, are inherently susceptible to environmental noise (e.g., multipath reflections, RF interference) and motion artifacts. Subtle biomarkers of dehydration, such as skin permittivity changes or capillary flow perturbations, are often obscured by subject movements that exceed 0.2 m / s. The 10 GHz cmWave band of CN0566, although advantageous for tissue penetration, exacerbates the sensitivity to dynamic clutter in uncontrolled environments.
- The WST's multi-layer decomposition, while generating stable features, imposes significant computational load. For example, processing 1-second radar frames (50 MSPS sampling) with three scattering layers requires around 10^6 multiplied accumulating operations, which challenges real-time deployment on edge devices.
- Current models (e.g., SVM, neural networks) exhibit performance degradation when applied to demographics

outside of training datasets, such as individuals with high BMI (>30) or elderly subjects with attenuated skin elasticity. This is due to the limited diversity of the data set and the inherent variability in the dynamics of subcutaneous hydration between populations.

- Correlating radar-derived hydration indices (e.g. normalized scattering energy) with gold-standard measures (e.g., serum osmolality, bioimpedance) remains incomplete. Discrepancies arise from temporal lags (>15 min) between physiological dehydration onset and detectable radar signatures, which complicates the real-time clinical utility.

2) *Future Research Directions*: We elaborate on the potential improvement directions for the project as follows:

- To mitigate computational bottlenecks, hardware-accelerated WST using FPGA-based modulus operators and parallel averaging pipelines will be explored. Furthermore, end-to-end learnable wavelet banks with dehydration labels could replace fixed filters, enhancing the relevance of the features.
- Federated learning frameworks will address the scarcity of data sets by aggregating anonymized data from distributed deployments. Domain adaptation techniques (e.g., adversarial training) will improve model generalization across demographics. Hybrid architectures, such as vision transformers, could capture long-range dependencies in time-frequency scattering coefficients.
- Augmenting radar data with thermal imaging (skin temperature) and optical photoplethysmography (blood volume pulses) will create a multiphysics hydration model. Sensor fusion via neural networks could dynamically weight modalities based on SNR conditions.
- Deploying quantized neural networks on low-power SoCs (e.g., Raspberry Pi 5 with NPU) will achieve <100 ms latency. Increasing the learning on the device will adapt the models to individual physiological baselines during the initial calibration phases.

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